

Codebook

Organizing Area	Theme	Code	Description	Example quote
Privacy Concerns and Perceived Threats	Unwanted and Malicious Content	Receipt of spam and unsolicited content	BLV users encounter spam, phishing links, and unwanted messages. Limited ability to visually verify content increases anxiety and reduces trust.	"When I get these random spam messages, I feel as my privacy is being violated." (P17)
		Risk of unintentional activation	Sequential focus, "dead areas," and unclear navigation lead to accidental activation of harmful links.	"You just have to take a chance to randomly tap on that dead area or double tap. . . which I don't like." (P19)
		Inability to assess link legitimacy	Participants cannot independently judge whether links from unknown senders are safe, relying on textual cues that are easy to misread or absent altogether.	"I'm scared to click the link because I think it's a virus or spyware link." (P13)
	Exposure of Personal or Sensitive Information by Default Settings	Inability to verify visible information	Participants cannot independently and quickly confirm what personal information is visible to others, which amplifies their general feeling of vulnerability.	"there are privacy settings, which may support all our privacy needs, but most of them may not be accessible" (P2) "You don't know if the privacy setting is turned on or off. . . it is not described in accessibility."(P13)
		Default group inclusion permissions	Concerns about being added to groups without consent due to permissive default settings, compounded by participants'	"I'd like if it[application] said, you know, hey, you've been asked to join this group, do you

			unawareness that controls to restrict this exist.	accept yes or no" (P1)
		Phone number visibility in group contexts	Default exposure of phone numbers in WhatsApp group chats, including to non-contacts, leaves participants with limited control over a disclosure they did not consent to.	"I don't want to be part of that group, and on top of that, my number is visible in that group" (P12)
		Profile and status indicator visibility	Concerns about default visibility of profile photos, last-seen timestamps, and online status, which participants address through deliberate withholding or disabling.	"I do not have a [profile] photo because I just don't want people to get my picture online" (P5)
		Auto-download as silent exposure	Default auto-download settings (in platforms such as WhatsApp and Telegram) store media on devices without user consent, creating hidden repositories that participants cannot easily monitor or manage and sometimes do not know about. Auto-download on recipients' devices also undermines later deletion.	"For it to be done automatically, I do feel like it's just whatever you send becomes the permanent property of the recipients" (P6)
	AI Distrust and Privacy Concerns	Distrust of AI data practices	Participants frame AI integration as entailing data collection, training, or surveillance over which they have limited control.	"If I enable AI, I know it's going to use what I share for training data" (P2)
		Reluctance to share sensitive content with AI	Participants selectively use AI tools and hesitate to share sensitive images with AI-based description services.	"They will probably record this picture because you already sent it to the AI. So I do feel more hesitant to send pictures with sensitive information" (P5)
		Disability bias in AI systems	Concerns that AI is trained primarily by sighted and non-disabled developers and	"Ninety percent of the time, AI is trained by non-disabled people,

			consequently misrepresents BLV experiences or applies biased interpretation.	so it starts acquiring its own disability biases" (P2)
Accessibility Barriers to Privacy	Lack of Awareness of Privacy Features	Limited knowledge of available features	Participants did not know about specific privacy and security controls (group permissions, read receipts, view-status, message timers, self-destructing messages) until prompted.	"I didn't know about the settings... I usually don't go to the settings unless I have a problem" (P6)
		Visual salience does not transfer to non-visual access	Visual salience cues that draw sighted users' attention to privacy controls (font weight, color, bolding, layout emphasis) have no equivalents in non-visual interaction, forcing linear discovery.	"Visually it might stand out by different font, different color, different bolding, or other indications that draw one's attention to it" (P11)
		Reliance on external sources for feature awareness	Participants learn about privacy features through external channels (peers, social networks, podcasts, web searches) rather than through the messaging application itself.	"Sometimes, you know, people in groups will talk about it, and they'll give advice on how to find it" (P18)
	Irreversible Disclosure Risk	Inability to verify content before sharing	Media pickers lack image descriptions, accessible metadata, and reliable selection feedback, making it unclear which image or file is about to be sent.	"You're never really sure if you're sending what you think you're sending" (P7)
		Inaccessible editing and deletion	Per-message edit and delete operations are difficult to perform reliably via screen reader, even when the equivalent chat-level operation is accessible.	"You can delete an entire chat easily and accessibly, but not individual messages" (P2)
		Time-sensitivity defeated by interaction speed	Screen-reader navigation delays mean that messages are typically read by recipients before BLV users can complete the deletion action.	"By the time I figure it out, the other person has already read it" (P7)

	Inaccessible Features Undermine Privacy and Security	Accessibility burden discourages feature adoption	Features that introduce additional steps, authentication, or navigation effort are perceived as impractical for routine messaging use.	"verification areas won't allow you to paste... there's a text box for each individual number... and the timer is super duper short." (P15)
		Opaque permission management	Participants accept broad app-level permissions to enable basic functionality but cannot later verify or confidently manage that access.	"It seems harder to know if it shares the location while using the app or not. Those settings are harder to access" (P11)
		Inaccessibility undermines technical security	Even strong technical security mechanisms (e.g., encryption) are functionally ineffective when their interfaces are inaccessible. Accessibility is articulated as a prerequisite for security.	"If you put something encrypted in front of me that is a WCAG failure nightmare, I don't care if it's encrypted... encryption is not enough if accessibility fails" (P2)
	Navigation and Discoverability	Privacy controls buried in menus	Privacy settings sit deep within nested menu structures, requiring extensive exploration that is particularly costly for screen reader users.	"A lot of times the privacy settings are buried in the back end, and so if you are not a person who goes digging around the back end, you're not gonna find them" (P2)
		Cross-platform inconsistency	Different messaging applications place privacy controls in different locations and use different navigation paths, forcing BLV users to memorize multiple inconsistent structures.	"I hit escape and oh if I hit escape here on the computer it's gonna take my whole chat box away... I think that's the only challenge, is just keeping up because so many are widely accepted across the country and in industry

				just keeping it all straight is the biggest challenge." (P1)
		Disruption from updates	Application updates relocate features and may reset privacy settings without notice, leaving participants uncertain whether their configurations remain intact.	"Once it's completely changed and something totally different, it's challenging and time consuming and frustrating" (P7)
		Unlabeled and ambiguous controls	Buttons and controls lack accessible labels, or labels are vague enough that the function cannot be inferred without trial.	"There are still a lot of like unlabeled buttons... it's oddly labeled... I just pushed stuff to try to figure out what it is" (P9, Discord)
		Effort as deterrent to privacy management	The time and effort required to find and configure privacy controls discourages participants from actively managing their privacy.	"Trying to change any [privacy] settings is like... you first need to find the place where that specific setting might be located, which sometimes is not easy to do... so it's pretty time-consuming" (P7)
Strategies to Protect Privacy in Instant Messaging Applications	Behavioral Risk Minimization	Limiting sensitive disclosure	Avoiding the sharing of high-sensitivity information (passwords, financial details, identification numbers) through messaging applications.	"I don't share, you know, passwords, and Social Security numbers" (P5)
		Adapting to ambient screen reader output	Modifying communication and device practices to account for the risk that screen reader output will be overheard by people nearby.	"The speech might read it out loud where everybody could hear... so I usually would have headphones on if I thought it was something I didn't want people to hear" (P19)

		Avoiding risky interaction	Refusing or limiting engagement with unknown senders, suspicious links, and ambiguous interface elements.	"You don't always want to visit links that you're not sure what they are" (P19)
		Channel switching for sensitive content	Moving conversations involving highly sensitive content (e.g., PINs, account numbers) to alternative channels considered more secure.	"If it needs to be a very private conversation, I'd rather have that, you know, face to face or over a landline, or you know, over a secure line" (P1)
	Securing Messaging Accounts and Devices	Strengthening authentication	Changing passwords and enabling additional security measures, often reactively in response to suspicious activity.	"When I noticed something unusual with my account, I just preferred to change my password and set secondary security measures" (P4)
		Selective permission management	Granting app-level permissions only when contextually justified by the app's function.	"I prefer not to [give access to share location with application]... otherwise I don't want the app to know where I am because there's no need for the app to know where I am" (P11)
		Physical device protection	Protecting the personal device itself, recognizing that messaging privacy is inseparable from device security.	"I do not engage them myself just because I am very, very protective of my devices" (P2)
	Leveraging App Features for Privacy Protection	Ephemeral content controls	Using features that allow temporary sharing or post-hoc removal of content to reduce the long-term footprint of sensitive disclosures.	"Sometimes there have been videos that I've sent as view once... I have used that feature before, and I also have sent and then deleted

				messages" (P11)
		Blocking and reporting	Using block and report functions to manage unwanted contact and reclaim a sense of control over the conversation surface.	"If I don't recognize the person... I just do like a report and block, or I can archive the chat on there" (P10)
		Group control without exit	Using mute and similar features to reduce disruption from groups without fully leaving them, particularly when groups were joined without consent.	"I will stay in there until you remove me. You added me, so I might mute it, but I might check in sometimes" (P11)